

NO TRIANGLE CAN BE CUT INTO NINETEEN CONGRUENT TRIANGLES: THE PRIME CASE OF ERDŐS PROBLEM 634

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ABSTRACT. We prove that for every prime $p > 3$ with $p \equiv 3 \pmod{4}$, no nondegenerate Euclidean triangle can be tiled by exactly p pairwise interior-disjoint congruent triangles. In particular, no triangle can be cut into nineteen congruent triangles, answering a question highlighted in Erdős Problem 634. Combined with the classical constructions realizing $p = 2$, $p = 3$, and every prime $p \equiv 1 \pmod{4}$, this determines the prime tile counts completely: a triangle can be tiled by a prime number p of congruent triangles if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod{4}$. The proof combines recent classification theorems of Beeson, Laczkovich, and Zhang with an elementary arithmetic argument excluding a prime tile count in the remaining configurations, in which the tile has a $2\pi/3$ angle. Several of the structural results we use are at present available only as preprints.

1. INTRODUCTION

Any triangle can be cut into n^2 congruent triangles similar to itself, for every $n \geq 1$. Erdős asked, in a question reported by Soifer [11], for which N there exists *some* triangle that can be cut into N congruent triangles; this is Problem 634 in Bloom’s collection [7]. Soifer showed that all N of the forms n^2 , $2n^2$, $3n^2$, $6n^2$, and $n^2 + m^2$ are realizable, and Zhang [12] recently gave further constructive families. In the other direction, Beeson proved that no triangle can be cut into 7 or into 11 congruent triangles [1], and in July 2026 that no *isosceles* triangle can be cut into a prime number $N > 3$ of congruent triangles [4]. The page [7] states: “It is possible that any prime of the form $4n + 3$ does not have this property. In particular, it is not known if 19 has this property.”

We settle this question.¹

Theorem 1. *Let $p > 3$ be a prime with $p \equiv 3 \pmod{4}$. No nondegenerate Euclidean triangle can be tiled by exactly p pairwise interior-disjoint congruent triangles. In particular, no triangle can be cut into 19 congruent triangles.*

Here a *tiling* of a triangle T by a triangle R is a finite family of triangles congruent to R (reflections allowed), with pairwise disjoint interiors, whose union is T ; the tiling is not assumed edge-to-edge. Following [5], a tiling is a *reptiling* if R is similar to T ; a triangle has *commensurable angles* if all of its angles are rational multiples of π , and *incommensurable angles* otherwise; and a triangle has *commensurable sides* if the pairwise ratios of its side

Date: July 24, 2026.

2020 Mathematics Subject Classification. Primary 52C20; Secondary 51M04, 11D09.

Key words and phrases. triangle tilings, congruent triangles, Erdős problem 634, prime tilings. ellamind GmbH, Bremen, Germany. *E-mail address:* jan@ellamind.com.

¹A withdrawn preprint of Beeson [2] had claimed, among other results, that no 19-tiling exists. It was withdrawn in 2024 after its main theorem was found to be false; the arXiv record notes that tilings described in later papers contradict it. The present proof makes no use of that manuscript.

lengths are rational. We call a triangle *scalene* if no two of its sides are equal; “isosceles” is used inclusively, so that equilateral triangles are isosceles and “not isosceles” means scalene.

Since every prime $p \equiv 1 \pmod{4}$ is a sum of two squares (Fermat), the classical hypotenuse construction of Snover, Waiveris, and Williams [10] realizes such p as tile counts for a right triangle, and 2 and 3 are realized by bisecting an isosceles right triangle and by the 3-tiling of the equilateral triangle. Theorem 1 is therefore sharp and yields:

Corollary 2. *A prime p is the tile count of some tiling of some triangle by congruent triangles if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod{4}$.*

This completes the prime case of Erdős Problem 634. In passing it recovers the cases $N = 7$ and $N = 11$, first proved by Beeson [1] by different methods.

The proof of Theorem 1 is short because it stands on recent structural results of Beeson, Laczkovich, and Zhang, recalled in Section 2, which reduce the problem to tiles with a $2\pi/3$ angle and commensurable sides. The remaining arithmetic input is Lemma 9 in Section 3. We do not claim novelty for its algebraic ingredients: the side vectors and tile-count expressions occurring there appear in Zhang’s constructions [12], and the basic arithmetic of the equation $c^2 = a^2 + ab + b^2$ is developed by Beeson in [4]; detailed attributions are given in Section 3. The point of Lemma 9 is the necessity statement needed here: in each of the four remaining configurations, for an *arbitrary* tiling, boundary integrality and primitivity force the scale factor between tile and tiled triangle to be an integer, and the resulting tile counts are never prime. Section 4 assembles the proof, and Section 5 describes supplementary computational checks of the algebra.

Theorem 1 relies on several recent results that are currently available only as preprints—[5, 6, 4] and, through Theorem 6 below, the manuscript [3]—and these have not all appeared in refereed venues at the time of writing. The precise statements used are reproduced in Section 2; we use them as black boxes and do not re-derive their hypotheses or conclusions.

2. IMPORTED RESULTS

Throughout, T denotes the tiled triangle and R the tile. We use the following results.

Theorem 3 (Laczkovich [8, 9]; stated as [5, Theorem 3]). *Suppose T has commensurable angles and is tiled by R . If T is not an isosceles (or equilateral) triangle, then the tiling is a reptiling.*

Theorem 4 (Snover–Waiveris–Williams [10]; stated as [5, Theorem 4]). *If T can be reptiled into N tiles, then $N = M^2$, or $N = M^2 + K^2$, or $N = 3M^2$ for some positive integers M, K .*

Theorem 5 (Beeson–Laczkovich–Zhang [5, Theorem 9], from Laczkovich [8, Theorem 4.1]). *Let a non-isosceles triangle T be tiled by a tile R with incommensurable angles (α, β, γ) . Then one of the following holds:*

- (1) *the tiling is a reptiling;*
- (2) (Group 1) $3\alpha + 2\beta = \pi$, and T has angles $(\alpha, 2\alpha, 2\beta)$ or $(2\alpha, \beta, \alpha + \beta)$;
- (3) (Group 2) $\gamma = 2\pi/3$, and T has angles

$$(\alpha, 2\alpha, 3\beta), \quad (\alpha, 2\beta, 2\alpha + \beta), \quad (\alpha, \alpha + \beta, \alpha + 2\beta), \quad \text{or} \quad (2\alpha, 2\beta, \alpha + \beta).$$

Theorem 6 (Beeson [4, Theorem 1], proved in [3]). *Let T be tiled by N copies of a tile R with angles (α, β, γ) satisfying $3\alpha + 2\beta = \pi$, where T is not similar to R . Then N is not prime.*

Theorem 7 (Beeson–Zhang [6, Theorem 1.2]; stated as [5, Theorem 6]). *Let T be tiled by a tile R such that R is not similar to T , R is not a right triangle, and R has incommensurable angles. Then R has commensurable sides.*

Theorem 8 (Beeson [4, Theorem 11]). *Let T be an isosceles triangle tiled into N copies of a triangle R . Then N is not a prime greater than three.*

In Theorem 5, the labeling of the two acute tile angles α, β in Group 2 is not ordered, so the four listed triples are to be read up to the interchange $\alpha \leftrightarrow \beta$. Lemma 9 below is symmetric under this interchange, and the computational checks of Section 5 run both labelings.

3. AN ARITHMETIC LEMMA FOR TILES WITH A $2\pi/3$ ANGLE

Lemma 9. *Let R have angles (α, β, γ) with $\gamma = 2\pi/3$ and commensurable sides, scaled so that the sides opposite (α, β, γ) are positive integers (a, b, c) with $\gcd(a, b, c) = 1$; we call such a triple primitive. If T has one of the four Group 2 angle triples of Theorem 5 and is tiled by N copies of R , then N is not prime.*

Proof. The law of cosines with $\gamma = 2\pi/3$ gives

$$(1) \quad c^2 = a^2 + ab + b^2.$$

Primitivity implies pairwise coprimality: a prime dividing two of a, b, c divides the third by (1) (cf. [4, Lemma 5]). Moreover

$$(2) \quad 3 \nmid c:$$

if $3 \mid c$, then (1) modulo 3 gives $(a - b)^2 \equiv 0 \pmod{3}$, so $a = b + 3k$; substituting, $c^2 = 3b^2 + 9bk + 9k^2$, and $9 \mid c^2$ forces $3 \mid b$ and then $3 \mid a$, contradicting primitivity (cf. the proof of [4, Theorem 6]).

Boundary integrality. With this scaling, every side of T has integer length; no edge-to-edge assumption is needed. Let S be a side of T and let ℓ be the line containing S . Every tile is contained in T , hence in the closed half-plane bounded by ℓ that contains T ; thus ℓ supports each tile it meets. If a tile meets the relative interior of S in more than one point, its intersection with ℓ is a face of the tile—a vertex or an entire edge—and, containing two points, it is an entire edge, of length a , b , or c . Two such edges have disjoint relative interiors: if the edges of two tiles overlapped in a nondegenerate segment of S , the two tiles would occupy a common half-neighborhood of an interior point of the overlap on the inner side of ℓ , contradicting disjoint interiors. The tiling is finite and covers S , and only finitely many points of S lie solely in tiles meeting S in a single point; hence S is partitioned, up to finitely many points, into complete tile edges, and $|S|$ is a sum of elements of $\{a, b, c\}$ —an integer.

Side and area computations. Let $K = 2c/\sqrt{3}$, the circumdiameter of R . The laws of sines and cosines give

$$K \sin \alpha = a, \quad K \sin \beta = b, \quad K \sin \gamma = c, \quad \cos \alpha = \frac{a + 2b}{2c}, \quad \cos \beta = \frac{2a + b}{2c}.$$

Since $\alpha + \beta = \pi/3$, the addition formulas yield

$$(3) \quad \begin{aligned} K \sin(\alpha + \beta) &= c, & K \sin 2\alpha &= \frac{a(a + 2b)}{c}, & K \sin 2\beta &= \frac{b(2a + b)}{c}, \\ K \sin(\alpha + 2\beta) &= K \sin(2\alpha + \beta) = a + b, & K \sin 3\beta &= \frac{3ab(a + b)}{c^2}. \end{aligned}$$

For a triangle with angles (A, B, C) and sine-law scale K_T (so its sides are $K_T \sin A$, etc.), the area equals $\frac{1}{2} K_T^2 \sin A \sin B \sin C$. Writing $m > 0$ for the scale factor from the displayed integer vector to the actual sides of T , formulas (3) give, for the four Group 2 cases:

Case	Angles of T	Integer side vector of T	Tile count N
A	$(\alpha, 2\alpha, 3\beta)$	$(c^2, c(a+2b), 3b(a+b))$	$3m^2(a+2b)(a+b)$
B	$(\alpha, 2\beta, 2\alpha+\beta)$	$(ac, b(2a+b), c(a+b))$	$m^2(2a+b)(a+b)$
C	$(\alpha, \alpha+\beta, \alpha+2\beta)$	$(a, c, a+b)$	$m^2(a+b)/b$
D	$(2\alpha, 2\beta, \alpha+\beta)$	$(a(a+2b), b(2a+b), c^2)$	$m^2(a+2b)(2a+b)$

Here N is the ratio of areas, computed from the sine products and the ratio of sine-law scales; the scale ratios K_T/K are mc^2/a , mc , m , and mc in cases A–D respectively. Primitivity of the four vectors is proved below.

Side vectors and tile counts equivalent to the rows of this table appear in Zhang’s paper [12], where they arise from explicit constructions: for case D in [12, Section 4] (Proposition 6), and for cases B, C, and A in [12, Propositions 10, 9, and 11], respectively. Moreover, for case D, Zhang’s Lemma 5 already proves the necessity statement we are after, under the hypothesis $a \not\equiv b \pmod{3}$ —which holds for every primitive triple, since $c^2 \equiv (a-b)^2 \pmod{3}$ and $3 \nmid c$ by (2). What the present lemma adds is the same necessity statement in all four cases, together with the exclusion of a prime count.

The vectors are primitive. In case C, $\gcd(a, c) = 1$. In case B, a common prime divisor q of all three entries satisfies $q \mid c$ (if $q \nmid c$ then $q \mid a$ and $q \mid a+b$, so $q \mid b$, impossible); then $q \nmid b$, so $q \mid 2a+b$, and reducing (1) modulo q using $b \equiv -2a$ gives $3a^2 \equiv 0 \pmod{q}$, so $q = 3$, contradicting (2). In case D, a common prime q divides c^2 but neither a nor b , hence divides $a+2b$ and $2a+b$, hence $2(a+2b) - (2a+b) = 3b$; since $q \nmid b$, this forces $q = 3$, again contradicting (2). In case A, a common prime q divides c , hence not $3b$ by (2) and coprimality, hence $q \mid a+b$; reducing (1) modulo q with $a \equiv -b$ gives $b^2 \equiv 0 \pmod{q}$, contradicting $\gcd(b, c) = 1$.

The scale factor is an integer. The sides of T are integers and equal m times the primitive vector $v = (v_1, v_2, v_3)$ of the relevant row. By Bézout, there are integers r_1, r_2, r_3 with $r_1 v_1 + r_2 v_2 + r_3 v_3 = 1$. Each mv_i is an integer, being a side length of T ; therefore

$$m = r_1(mv_1) + r_2(mv_2) + r_3(mv_3) \in \mathbb{Z},$$

and $m > 0$ gives $m \in \mathbb{Z}_{>0}$.

Conclusion. Since $a, b \geq 1$, we have $a+b \geq 2$, $a+2b \geq 3$, and $2a+b \geq 3$, while $m \geq 1$. In cases A, B, and D the count N is therefore a product of two integer factors each greater than one—for instance $N = (m^2(2a+b)) \cdot (a+b)$ in case B—hence composite (in case A, N is moreover divisible by 3).

In case C, suppose $N = p$ is prime. Then $pb = m^2(a+b)$. Since $\gcd(b, a+b) = \gcd(a, b) = 1$, we get $b \mid m^2$; write $m^2 = br$ with $r \in \mathbb{Z}_{>0}$. Then $p = r(a+b)$, and since p is prime and $a+b \geq 2$, necessarily $r = 1$ and

$$p = a+b.$$

Now (1) gives $c^2 = (a+b)^2 - ab = p^2 - ab$, so $0 < c < p$. Modulo p we have $b \equiv -a$, hence $c^2 \equiv -ab \equiv a^2 \pmod{p}$, so $p \mid (c-a)(c+a)$. Since $0 < a, c < p$, either $c = a$ or $c+a = p$, i.e. $c = b$. But $c = a$ in (1) forces $b(a+b) = 0$ and $c = b$ forces $a(a+b) = 0$, both impossible. Hence N is not prime in case C either.

All four cases (and, by the $a \leftrightarrow b$ symmetry of the argument, their relabelings) are settled. $\square$

4. PROOF OF THEOREM 1

Proof. Let $p > 3$ be prime with $p \equiv 3 \pmod{4}$ and suppose a triangle T is tiled by p congruent copies of a tile R .

Step 1: T is not isosceles. This is Theorem 8, since $p > 3$. In particular T is scalene.

Step 2: the tiling is not a reptiling. Otherwise Theorem 4 gives $p = M^2$, $p = M^2 + K^2$, or $p = 3M^2$. A prime is not a square. If $p = 3M^2$, then $3 \mid p$, so $p = 3$ and $M = 1$, contrary to $p > 3$. And $p \equiv 3 \pmod{4}$ is not a sum of two squares, since squares are 0 or 1 modulo 4. In particular R is not similar to T .

Step 3: R has incommensurable angles. Each corner of T has angle less than π . The corner point cannot lie in the interior of a tile, where the tile would contribute angle 2π , nor in the relative interior of a tile edge, where it would contribute angle π ; hence it is a vertex of every tile containing it, and each angle of T is a finite sum of tile angles. If all angles of R were rational multiples of π , the same would hold for T ; then Theorem 3 (T being scalene) would force a reptiling, contradicting Step 2.

Step 4: the two remaining groups. By Theorem 5 and Step 2, the tiling belongs to Group 1 or Group 2. In Group 1, Theorem 6 applies (T is not similar to R by Step 2) and shows $N = p$ is not prime, a contradiction. In Group 2, the tile satisfies $\gamma = 2\pi/3$, so $\alpha + \beta = \pi/3$ and $0 < \alpha, \beta < \pi/3$; in particular R is not a right triangle. Since R is not similar to T and has incommensurable angles, Theorem 7 shows R has commensurable sides, and Lemma 9 shows $N = p$ is not prime, again a contradiction.

All cases are impossible, so no such tiling exists. Taking $p = 19$ gives the titular statement. $\square$

5. COMPUTATIONAL CHECKS

The proofs of all assertions in this note are contained in the text; the computations described here are supplementary checks. An accompanying Python script uses exact symbolic arithmetic (SymPy) throughout, with no floating-point computation; see the ancillary files `verify_triangle19.py` and `README.txt`.

Exact symbolic checks. The script verifies the six trigonometric identities (3) and the four area-ratio formulas of the table symbolically, modulo the relation (1) and in both labelings of the acute angles.

Finite stress tests. The primitivity (gcd) assertions for the four side vectors are universal statements, proved above; the script additionally tests them on 41,112 primitive solutions of (1), generated in both labelings from the two-branch parametrization of [4, Theorem 6] over $2 \leq s \leq 300$, $1 \leq t < s$, $\gcd(s, t) = 1$, $s \not\equiv t \pmod{3}$. Such finite testing serves as error detection and is not a proof. As a further audit of case C for $p = 19$, the script checks that all four candidate pairs $(a, b) = (19 - m^2, m^2)$, $1 \leq m \leq 4$, fail to make $c^2 = a^2 + ab + b^2$ a perfect square. It also verifies the identity $a + b = s(s + 2t)$ on the parametrization, which yields a second route to the case C conclusion, since $s \geq 2$ and $s + 2t \geq 4$ make $a + b$ composite.

USE OF AI TOOLS

This note was produced with substantial use of AI systems, which we disclose in the interest of transparency. An initial version of Lemma 9, of the assembly in Section 4, and of the verification script was generated by OpenAI GPT-5.6 Pro on July 24, 2026. Claude

(Anthropic, Fable 5) was used as a second-model cross-check: it re-derived the identities and arguments of Lemma 9 and Section 4, compared the statements of the imported theorems against the cited arXiv versions, and audited and executed the verification script. Cross-checking by a second model is a useful error filter, but it is not independent verification in the scholarly sense, and neither the model-based checks nor the computations of Section 5 replace the proofs given in the text. The author reviewed the manuscript and takes full responsibility for its content. The imported Theorems 3–8 are human mathematics due to the cited authors; no claim of AI contribution is made for them.

ACKNOWLEDGEMENTS

This note builds directly on the program of Michael Beeson, Miklós Laczkovich, and Yan X. Zhang, whose structural theorems carry the bulk of the mathematical weight, and on Thomas Bloom’s curation of the Erdős problems collection.

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